

1 Kronecker-Separable Spatio-Temporal HiPPO-SVGP

1.1 Problem Setup

We consider a spatio-temporal latent function

$$f : \mathcal{S} \times \mathbb{R} \rightarrow \mathbb{R}, \quad (1)$$

where $s \in \mathcal{S} \subset \mathbb{R}^{d_s}$ denotes spatial location and $t \in \mathbb{R}$ denotes time. The observations are

$$y_{n,i} = y(t_n, s_i), \quad n = 1, \dots, N_t, \quad i = 1, \dots, N_s. \quad (2)$$

We assume a Gaussian process prior over f ,

$$f \sim \mathcal{GP}(0, k), \quad (3)$$

with a separable spatio-temporal kernel

$$k((s, t), (s', t')) = k_s(s, s') k_t(t, t'). \quad (4)$$

Let

$$K_t \in \mathbb{R}^{N_t \times N_t}, \quad (K_t)_{nn'} = k_t(t_n, t_{n'}), \quad (5)$$

and

$$K_s \in \mathbb{R}^{N_s \times N_s}, \quad (K_s)_{ii'} = k_s(s_i, s_{i'}). \quad (6)$$

If the latent values are vectorized as

$$\mathbf{f} = \text{vec}([f(t_n, s_i)]_{n,i}) \in \mathbb{R}^{N_t N_s}, \quad (7)$$

then the full prior covariance factorizes as

$$K_{ff} = K_t \otimes K_s. \quad (8)$$

For Gaussian observations, we assume

$$y \mid f \sim \mathcal{N}(f, \sigma^2 I), \quad (9)$$

where $y \in \mathbb{R}^{N_t N_s}$ is the vectorized observation vector.

1.2 Temporal HiPPO-RFF Interdomain Features

On the temporal axis, we employ HiPPO-LegS interdomain inducing variables together with a random Fourier feature (RFF) approximation of the stationary temporal kernel.

1.2.1 HiPPO-LegS Basis

For each time horizon $t > 0$, define the scaled Legendre basis

$$g_\ell^{(t)}(x) = \sqrt{\frac{2\ell+1}{t}} P_\ell\left(\frac{2x}{t} - 1\right), \quad x \in [0, t], \quad (10)$$

where P_ℓ is the Legendre polynomial of degree ℓ .

The temporal interdomain inducing variables are defined as

$$u_\ell(t) = \int_0^t g_\ell^{(t)}(x) f_t(x) dx, \quad \ell = 0, \dots, M_t - 1. \quad (11)$$

1.2.2 RFF Approximation of the Temporal Kernel

Assume the temporal kernel k_t is stationary. Using Bochner's theorem, we approximate it via random Fourier features:

$$k_t(\tau, \tau') \approx \sum_{r=1}^R w_r \left[\cos(\omega_r \tau) \cos(\omega_r \tau') + \sin(\omega_r \tau) \sin(\omega_r \tau') \right]. \quad (12)$$

Here ω_r are frequencies sampled from the spectral density of k_t , and w_r are the corresponding quadrature or Monte Carlo weights.

1.2.3 Closed-Form HiPPO-RFF Integrals

Define the oscillatory integrals

$$I_\ell^{\cos}(t; \omega) := \int_0^t \cos(\omega x) g_\ell^{(t)}(x) dx, \quad (13)$$

$$I_\ell^{\sin}(t; \omega) := \int_0^t \sin(\omega x) g_\ell^{(t)}(x) dx. \quad (14)$$

Using the change of variables $u = 2x/t - 1$ and the spherical-Bessel identity

$$\int_{-1}^1 e^{i\kappa u} P_\ell(u) du = 2i^\ell j_\ell(\kappa), \quad \kappa = \frac{\omega t}{2}, \quad (15)$$

these integrals admit the closed forms

$$I_\ell^{\cos}(t; \omega) = \sqrt{(2\ell+1)t} j_\ell(\kappa) \cos\left(\omega t_0 + \kappa + \frac{\ell\pi}{2}\right), \quad (16)$$

$$I_\ell^{\sin}(t; \omega) = \sqrt{(2\ell+1)t} j_\ell(\kappa) \sin\left(\omega t_0 + \kappa + \frac{\ell\pi}{2}\right), \quad (17)$$

where $j_\ell(\cdot)$ denotes the spherical Bessel function of the first kind, and t_0 denotes the global phase origin (`phase_origin`). Hence, under the `global_start` setting, the phase is $\omega t_0 + \kappa + \ell\pi/2$; under the `zero` branch, $t_0 = 0$, which recovers the original formula.

1.2.4 Temporal Inducing Covariances

Let

$$u^{(t)}(t) = (u_0(t), u_1(t), \dots, u_{M_t-1}(t))^\top. \quad (18)$$

Then the temporal inducing covariance matrix is

$$[K_{uu}^{(t)}(t)]_{\ell m} \approx \sum_{r=1}^R w_r \left[I_\ell^{\cos}(t; \omega_r) I_m^{\cos}(t; \omega_r) + I_\ell^{\sin}(t; \omega_r) I_m^{\sin}(t; \omega_r) \right]. \quad (19)$$

The point-to-inducing temporal cross-covariance is

$$[k_{fu}^{(t)}(\tau; t)]_\ell \approx \sum_{r=1}^R w_r \left[\cos(\omega_r \tau) I_\ell^{\cos}(t; \omega_r) + \sin(\omega_r \tau) I_\ell^{\sin}(t; \omega_r) \right]. \quad (20)$$

Substituting (16)–(17) yields

$$[k_{fu}^{(t)}(\tau; t)]_\ell \approx \sum_{r=1}^R w_r \sqrt{(2\ell+1)t} j_\ell(\kappa_r) \cos\left(\omega_r \tau - \omega_r t_0 - \kappa_r - \frac{\ell\pi}{2}\right). \quad (21)$$

For compactness, define the feature matrix

$$B_t(t) \in \mathbb{R}^{M_t \times 2R}, \quad (22)$$

with entries

$$[B_t(t)]_{\ell, r} = \sqrt{w_r} I_\ell^{\cos}(t; \omega_r), \quad [B_t(t)]_{\ell, R+r} = \sqrt{w_r} I_\ell^{\sin}(t; \omega_r), \quad (23)$$

and the RFF vector

$$\phi_t(\tau) = (\sqrt{w_1} \cos(\omega_1 \tau), \dots, \sqrt{w_R} \cos(\omega_R \tau), \sqrt{w_1} \sin(\omega_1 \tau), \dots, \sqrt{w_R} \sin(\omega_R \tau))^\top. \quad (24)$$

Then

$$K_{uu}^{(t)}(t) \approx B_t(t) B_t(t)^\top, \quad (25)$$

$$k_{fu}^{(t)}(\tau; t) \approx \phi_t(\tau)^\top B_t(t)^\top. \quad (26)$$

1.2.5 Local Temporal Blocks for Mini-Batch Training

For implementation, it is convenient to introduce local temporal batches

$$\mathcal{B}_b = \{(t_n, s_i) : t_n \in [T_b^{\text{start}}, T_b^{\text{end}}]\}, \quad b = 1, \dots, B, \quad (27)$$

and to associate each batch with a local horizon $t^{(b)}$. The same HiPPO-RFF construction is then applied batchwise by replacing t with $t^{(b)}$, yielding local temporal feature matrices $B_t^{(b)}$ and local temporal cross-covariances $K_{fu}^{(t,b)}$. This modification does not change the model itself; rather, it provides an implementation-oriented view that supports stochastic optimization over time while preserving the same analytic HiPPO-RFF structure.

1.3 Spatial Inducing Variables

On the spatial axis, we use standard inducing locations

$$Z_s = \{z_1^{(s)}, \dots, z_{M_s}^{(s)}\}. \quad (28)$$

Define the spatial inducing covariance

$$K_{ZZ}^{(s)} \in \mathbb{R}^{M_s \times M_s}, \quad (K_{ZZ}^{(s)})_{jj'} = k_s(z_j^{(s)}, z_{j'}^{(s)}), \quad (29)$$

and the spatial cross-covariance matrix

$$K_{XZ}^{(s)} \in \mathbb{R}^{N_s \times M_s}, \quad (K_{XZ}^{(s)})_{ij} = k_s(s_i, z_j^{(s)}). \quad (30)$$

1.4 Mixed Spatio-Temporal Inducing Variables

For each spatial inducing location $z_j^{(s)}$, define a temporal inducing vector

$$u_j(t) \in \mathbb{R}^{M_t}, \quad [u_j(t)]_\ell = \int_0^t g_\ell^{(t)}(x) f(x, z_j^{(s)}) dx. \quad (31)$$

Stacking all spatial inducing locations yields the mixed spatio-temporal inducing vector

$$u(t) = \text{vec}([u_1(t), u_2(t), \dots, u_{M_s}(t)]) \in \mathbb{R}^{M_t M_s}. \quad (32)$$

1.5 Kronecker Structure of the Mixed Inducing Covariances

By separability of the spatio-temporal kernel, the covariance between two mixed inducing entries factorizes into a temporal part and a spatial part. In particular, for temporal indices ℓ, m and spatial inducing locations $z_j^{(s)}, z_{j'}^{(s)}$, we have

$$\text{cov}(u_{\ell,j}(t), u_{m,j'}(t)) = [K_{uu}^{(t)}(t)]_{\ell m} [K_{ZZ}^{(s)}]_{jj'}. \quad (33)$$

Therefore, the full mixed inducing covariance matrix is

$$K_{uu}^{(st)}(t) = K_{uu}^{(t)}(t) \otimes K_{ZZ}^{(s)}. \quad (34)$$

Substituting the analytic temporal expression (25), we further obtain

$$K_{uu}^{(st)}(t) \approx (B_t(t) B_t(t)^\top) \otimes K_{ZZ}^{(s)}. \quad (35)$$

Similarly, for any data point (t_n, s_i) , the cross-covariance between $f(t_n, s_i)$ and the mixed inducing vector $u(t)$ factorizes as

$$K_{fu}^{(st)}((t_n, s_i); t) = k_{fu}^{(t)}(t_n; t) \otimes k_{sZ}^{(s)}(s_i). \quad (36)$$

Substituting the analytic temporal form (26), this becomes

$$K_{fu}^{(st)}((t_n, s_i); t) \approx (\phi_t(t_n)^\top B_t(t)^\top) \otimes k_{sZ}^{(s)}(s_i). \quad (37)$$

If we stack all time points and all spatial locations, then the overall point-to-inducing covariance matrix can be written as

$$K_{fu}^{(st)}(t) = K_{fu}^{(t)}(t) \otimes K_{XZ}^{(s)}, \quad (38)$$

where

$$K_{fu}^{(t)}(t) = \begin{bmatrix} k_{fu}^{(t)}(t_1; t) \\ \vdots \\ k_{fu}^{(t)}(t_{N_t}; t) \end{bmatrix} \in \mathbb{R}^{N_t \times M_t}, \quad (39)$$

and

$$K_{XZ}^{(s)} \in \mathbb{R}^{N_s \times M_s}, \quad (K_{XZ}^{(s)})_{ij} = k_s(s_i, z_j^{(s)}). \quad (40)$$

For a local temporal batch $\mathcal{B}_b$, the same factorization is used in the form

$$K_{fu}^{(b)} = K_{fu}^{(t,b)} \otimes K_{XZ}^{(s)}, \quad (41)$$

where $K_{fu}^{(t,b)}$ denotes the temporal cross-covariance matrix evaluated only at the query times contained in $\mathcal{B}_b$. This batchwise form will be useful for stochastic optimization.

In what follows, we use the superscript (st) to emphasize the full spatio-temporal quantities when needed. When no ambiguity arises, we may drop the superscript (st) for notational simplicity.

1.6 Conditional Prior $p(f | u)$

Using the mixed spatio-temporal inducing variables defined above, the standard Gaussian conditioning formula gives

$$p(f | u) = \mathcal{N}(f | K_{fu}^{(st)}(t) (K_{uu}^{(st)}(t))^{-1} u, K_{ff}^{(st)} - K_{fu}^{(st)}(t) (K_{uu}^{(st)}(t))^{-1} K_{uf}^{(st)}(t)). \quad (42)$$

Here the full spatio-temporal prior covariance is

$$K_{ff}^{(st)} = K_t \otimes K_s. \quad (43)$$

Using (34) and (38), together with

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}, \quad (A \otimes B)(C \otimes D) = (AC) \otimes (BD), \quad (44)$$

we obtain

$$K_{fu}^{(st)}(t) (K_{uu}^{(st)}(t))^{-1} = \left(K_{fu}^{(t)}(t) (K_{uu}^{(t)}(t))^{-1} \right) \otimes \left(K_{XZ}^{(s)} (K_{ZZ}^{(s)})^{-1} \right). \quad (45)$$

Hence the conditional mean is

$$m_{f|u} = \left(K_{fu}^{(t)}(t) (K_{uu}^{(t)}(t))^{-1} \otimes K_{XZ}^{(s)} (K_{ZZ}^{(s)})^{-1} \right) u. \quad (46)$$

Similarly, the projection term in the conditional covariance can be written as

$$K_{fu}^{(st)}(t) (K_{uu}^{(st)}(t))^{-1} K_{uf}^{(st)}(t) = \left(K_{fu}^{(t)}(t) (K_{uu}^{(t)}(t))^{-1} K_{uf}^{(t)}(t) \right) \otimes \left(K_{XZ}^{(s)} (K_{ZZ}^{(s)})^{-1} K_{ZX}^{(s)} \right). \quad (47)$$

Define the temporal and spatial projection matrices

$$P_t := K_{fu}^{(t)}(t)(K_{uu}^{(t)}(t))^{-1}K_{uf}^{(t)}(t), \quad P_s := K_{XZ}^{(s)}(K_{ZZ}^{(s)})^{-1}K_{ZX}^{(s)}. \quad (48)$$

Then the conditional covariance becomes

$$\Sigma_{f|u} = K_t \otimes K_s - P_t \otimes P_s. \quad (49)$$

Define the residual matrices

$$R_t := K_t - P_t, \quad R_s := K_s - P_s. \quad (50)$$

Using distributivity,

$$K_t \otimes K_s - P_t \otimes P_s = K_t \otimes R_s + R_t \otimes P_s. \quad (51)$$

Therefore,

$$\Sigma_{f|u} = K_t \otimes R_s + R_t \otimes P_s. \quad (52)$$

Equation (52) makes explicit the two approximation errors: the first term corresponds to *spatial inducing approximation error*, while the second term captures *temporal inducing approximation error*. When no ambiguity arises, we shall drop the superscript (*st*) and write simply K_{uu} , K_{fu} , and K_{ff} .

1.7 Variational Posterior

We use the standard Gaussian variational family

$$q(u) = \mathcal{N}(u \mid m, S). \quad (53)$$

Then

$$q(f, u) = p(f \mid u) q(u), \quad q(f) = \int p(f \mid u) q(u) du. \quad (54)$$

Since everything is Gaussian,

$$q(f) = \mathcal{N}(f \mid \mu_f, \Sigma_f), \quad (55)$$

with

$$\mu_f = Am, \quad A := K_{fu}K_{uu}^{-1}, \quad (56)$$

and

$$\Sigma_f = \Sigma_{f|u} + ASA^\top. \quad (57)$$

In the most general formulation, S is unrestricted. For implementation, one may optionally adopt a Kronecker-aware parameterization, such as $S \approx S_t \otimes S_s$ (Kronecker posterior), or a diagonal/low-rank approximation (e.g. $q(u) = \mathcal{N}(u \mid m, \text{diag}(s))$ or $S \approx \text{diag}(s) + UU^\top$), in order to better align the variational family with the Kronecker structure of the model. This does not alter the derivations below, which remain valid for the full Gaussian posterior.

1.8 Gaussian-Likelihood ELBO

Under the Gaussian likelihood (9), the standard sparse variational ELBO is

$$\mathcal{L} = \mathbb{E}_{q(f)}[\log p(y \mid f)] - \text{KL}[q(u) \parallel p(u)]. \quad (58)$$

The expected log-likelihood term is

$$\mathbb{E}_{q(f)}[\log p(y \mid f)] = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \left[(y - \mu_f)^\top (y - \mu_f) + \text{tr}(\Sigma_f) \right]. \quad (59)$$

Substituting (56) and (57),

$$\mathbb{E}_{q(f)}[\log p(y \mid f)] = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \left[(y - Am)^\top (y - Am) + \text{tr}(\Sigma_{f|u}) + \text{tr}(A^\top A S) \right]. \quad (60)$$

The KL divergence between $q(u) = \mathcal{N}(m, S)$ and $p(u) = \mathcal{N}(0, K_{uu})$ is

$$\text{KL}[q(u) \parallel p(u)] = \frac{1}{2} \left[\text{tr}(K_{uu}^{-1} S) + m^\top K_{uu}^{-1} m - \log |S| + \log |K_{uu}| - M \right]. \quad (61)$$

Combining (60) and (61), we obtain the closed-form ELBO:

$$\boxed{\begin{aligned} \mathcal{L} = & -\frac{N}{2} \log(2\pi\sigma^2) \\ & -\frac{1}{2\sigma^2} \left[(y - Am)^\top (y - Am) + \text{tr}(\Sigma_{f|u}) + \text{tr}(A^\top A S) \right] \\ & -\frac{1}{2} \left[\text{tr}(K_{uu}^{-1} S) + m^\top K_{uu}^{-1} m - \log |S| + \log |K_{uu}| - M \right]. \end{aligned}} \quad (62)$$

For stochastic optimization with local temporal batches, one may equivalently view the

expected log-likelihood as a sum of batch contributions,

$$\mathcal{L} = \sum_{b=1}^B \mathbb{E}_{q(f_{\mathcal{B}_b})} [\log p(y_{\mathcal{B}_b} \mid f_{\mathcal{B}_b})] - \text{KL}[q(u) \parallel p(u)], \quad (63)$$

where each batch distribution $q(f_{\mathcal{B}_b})$ is constructed using the local cross-covariance in (41). Equation (63) is an implementation-oriented reformulation of the same sparse variational objective.

1.9 Gradients with Respect to m and S

Let

$$B := A^\top A. \quad (64)$$

Then the gradient of (62) with respect to m is

$$\frac{\partial \mathcal{L}}{\partial m} = \frac{1}{\sigma^2} A^\top (y - Am) - K_{uu}^{-1} m. \quad (65)$$

The gradient with respect to S is

$$\frac{\partial \mathcal{L}}{\partial S} = -\frac{1}{2\sigma^2} A^\top A - \frac{1}{2} K_{uu}^{-1} + \frac{1}{2} S^{-1}. \quad (66)$$

1.10 Closed-Form Optimum Under Gaussian Likelihood

Since the Gaussian likelihood is conjugate, the optimal covariance is obtained by setting (66) to zero:

$$S^{-1} = K_{uu}^{-1} + \frac{1}{\sigma^2} A^\top A. \quad (67)$$

Hence

$$S = \left(K_{uu}^{-1} + \frac{1}{\sigma^2} A^\top A \right)^{-1}. \quad (68)$$

Similarly, setting (65) to zero yields

$$(A^\top A + \sigma^2 K_{uu}^{-1}) m = A^\top y, \quad (69)$$

and therefore

$$m = (A^\top A + \sigma^2 K_{uu}^{-1})^{-1} A^\top y. \quad (70)$$

Equivalently,

$$m = S \frac{1}{\sigma^2} A^\top y. \quad (71)$$

1.11 Natural Gradient Formulation

For the Gaussian variational posterior $q(u) = \mathcal{N}(m, S)$, the natural parameters are

$$\lambda^{(1)} = S^{-1}m, \quad \lambda^{(2)} = -\frac{1}{2}S^{-1}. \quad (72)$$

The Gaussian-likelihood optimum implies

$$S^{-1} = K_{uu}^{-1} + \frac{1}{\sigma^2}A^\top A, \quad S^{-1}m = \frac{1}{\sigma^2}A^\top y. \quad (73)$$

Therefore, the target natural parameters are

$$\boxed{\lambda_\star^{(2)} = -\frac{1}{2} \left(K_{uu}^{-1} + \frac{1}{\sigma^2}A^\top A \right)}, \quad (74)$$

$$\boxed{\lambda_\star^{(1)} = \frac{1}{\sigma^2}A^\top y}. \quad (75)$$

A damped natural gradient update may therefore be written as

$$\lambda_{\text{new}}^{(1)} = (1 - \beta)\lambda^{(1)} + \beta \frac{1}{\sigma^2}A^\top y, \quad (76)$$

$$\lambda_{\text{new}}^{(2)} = (1 - \beta)\lambda^{(2)} + \beta \left[-\frac{1}{2} \left(K_{uu}^{-1} + \frac{1}{\sigma^2}A^\top A \right) \right]. \quad (77)$$

The moment parameters are then recovered via

$$S = (-2\lambda^{(2)})^{-1}, \quad m = S\lambda^{(1)}. \quad (78)$$

1.12 CVI / Approximate-Likelihood Interpretation

Following the conjugate-computation variational inference (CVI) viewpoint, the variational posterior over inducing variables may be written as

$$q(u) \propto \mathcal{N}(Y_u^e \mid u, V_u^e) p(u), \quad (79)$$

where the Gaussian approximate likelihood parameters are related to the likelihood contribution to the natural parameters by

$$V_u^e = (-2\lambda_e^{(2)})^{-1}, \quad Y_u^e = V_u^e \lambda_e^{(1)}. \quad (80)$$

Under the Gaussian likelihood, the likelihood contribution to the natural parameters is

$$\lambda_e^{(2)} = -\frac{1}{2\sigma^2}A^\top A, \quad \lambda_e^{(1)} = \frac{1}{\sigma^2}A^\top y. \quad (81)$$

Hence

$$V_u^e = \sigma^2(A^\top A)^{-1}, \quad Y_u^e = (A^\top A)^{-1}A^\top y. \quad (82)$$

Thus the posterior admits the explicit CVI form

$$\boxed{q(u) \propto \mathcal{N}(Y_u^e \mid u, V_u^e) p(u).} \quad (83)$$

1.13 Online Spatio-Temporal Variational Updates

The preceding derivations describe a static variational posterior for the full spatio-temporal dataset. We now extend this formulation to a streaming setting in which observations arrive sequentially in temporal blocks. Our online construction follows three principles. First, data are assimilated batch by batch in time. Second, as time advances, only the temporal prior cross-covariance and temporal inducing covariance are updated, while the spatial inducing locations remain fixed. Third, after forming the updated Kronecker-separable spatio-temporal matrices, we perform a sequential variational posterior update using the newly arrived temporal block.

This is the setting of primary interest here: online updates apply only to the temporal dimension, whereas the spatial inducing set $Z_s = \{z_1^{(s)}, \dots, z_{M_s}^{(s)}\}$ is shared across time and is not updated online. This viewpoint is consistent with the online HiPPO-SVGP recursion on the temporal side and with the CVI / filtering interpretation used in spatio-temporal variational GP formulations.

Let the data arrive in temporal blocks

$$\mathcal{D}_n = \{(t_n, s_i, y_{n,i})\}_{i=1}^{N_s}, \quad n = 1, 2, \dots, \quad (84)$$

or, more generally, in mini-batches indexed by time. Denote by u_n the spatio-temporal inducing state associated with the temporal horizon up to time t_n . The online variational objective is to update the posterior summary sequentially,

$$q_{n-1}(u) \longrightarrow q_n(u), \quad (85)$$

using only the previous posterior and the newly arrived data block $\mathcal{D}_n$.

1.13.1 Sequential variational update

From the variational Bayes perspective, the posterior after observing the new block may be written in the unnormalised form

$$q_n(u) \propto q_{n-1}(u) \exp\left(\mathbb{E}_{p(f_n|u)}[\log p(y_n | f_n)]\right), \quad (86)$$

where f_n denotes the latent function values associated with the new time block, and y_n is the corresponding observation vector.

Equation (86) shows that the online posterior is obtained by multiplying the previous posterior summary with the likelihood contribution of the new temporal block. Thus the historical information is compressed into $q_{n-1}(u)$, and each new temporal batch contributes only an incremental update. Importantly, in the present separable construction, the spatial part of the inducing representation is fixed, and only the temporal part evolves with n .

1.13.2 Kronecker-separable structure of the online block update

Inspired by the online HiPPO-SVGP recursion, we regard the temporal quantities $K_{fu,n}^{(t)}$ and $K_{uu,n}^{(t)}$ as the time-evolving objects, whereas the spatial factors $K_{XZ}^{(s)}$ and $K_{ZZ}^{(s)}$ are fixed across all temporal blocks.

For the n -th temporal block, the mixed point-to-inducing covariance retains the same separable structure,

$$K_{fu,n}^{(st)} = K_{fu,n}^{(t)} \otimes K_{XZ}^{(s)}, \quad (87)$$

and the inducing covariance is

$$K_{uu,n}^{(st)} = K_{uu,n}^{(t)} \otimes K_{ZZ}^{(s)}. \quad (88)$$

Therefore the corresponding projection matrix becomes

$$A_n := K_{fu,n}^{(st)} (K_{uu,n}^{(st)})^{-1} = \left(K_{fu,n}^{(t)} (K_{uu,n}^{(t)})^{-1} \right) \otimes \left(K_{XZ}^{(s)} (K_{ZZ}^{(s)})^{-1} \right). \quad (89)$$

Equation (89) makes explicit that the spatial factor $K_{XZ}^{(s)} (K_{ZZ}^{(s)})^{-1}$ is shared across all time steps, while the temporal factor changes with the current time block. Hence, on-line adaptation occurs through the temporal component only, while the spatial inducing geometry remains unchanged.

1.13.3 Sequential Gaussian update in natural parameters

Suppose the online variational posterior at step n is

$$q_n(u) = \mathcal{N}(u \mid m_n, S_n), \quad (90)$$

with natural parameters

$$\lambda_n^{(1)} = S_n^{-1} m_n, \quad \lambda_n^{(2)} = -\frac{1}{2} S_n^{-1}. \quad (91)$$

Under the Gaussian likelihood, the contribution of the new time block is

$$\lambda_{e,n}^{(1)} = \frac{1}{\sigma^2} A_n^\top y_n, \quad \lambda_{e,n}^{(2)} = -\frac{1}{2\sigma^2} A_n^\top A_n. \quad (92)$$

Hence a sequential natural-gradient / CVI update can be written as

$$\lambda_n^{(1)} = (1 - \beta) \lambda_{n-1}^{(1)} + \beta \left(K_{uu,n}^{-1} m_{0,n} + \lambda_{e,n}^{(1)} \right), \quad (93)$$

$$\lambda_n^{(2)} = (1 - \beta) \lambda_{n-1}^{(2)} + \beta \left(-\frac{1}{2} K_{uu,n}^{-1} + \lambda_{e,n}^{(2)} \right), \quad (94)$$

where $m_{0,n}$ denotes the prior mean contribution at time n , which is zero in the present formulation. In the undamped case $\beta = 1$, this reduces to a full assimilation of the new block contribution.

Equivalently, the online posterior may be maintained through sufficient statistics. Define the posterior precision matrix and information vector as

$$\Lambda_n := S_n^{-1}, \quad h_n := S_n^{-1} m_n. \quad (95)$$

Then the recursive update takes the form

$$\Lambda_n = \Lambda_{n-1} + \frac{1}{\sigma^2} A_n^\top A_n, \quad h_n = h_{n-1} + \frac{1}{\sigma^2} A_n^\top y_n, \quad (96)$$

and the posterior moments are recovered by

$$S_n = \Lambda_n^{-1}, \quad m_n = S_n h_n. \quad (97)$$

For clarity, the same recursion may be written in unrolled form as

$$S_n^{-1} = \Lambda_0 + \sum_{j=1}^n \frac{1}{\sigma^2} A_j^\top A_j, \quad (98)$$

$$S_n^{-1}m_n = h_0 + \sum_{j=1}^n \frac{1}{\sigma^2} A_j^\top y_j, \quad (99)$$

where (Λ_0, h_0) denote the initial prior summary. In the zero-mean case, one may take $h_0 = 0$, while Λ_0 is induced by the initial prior over the mixed inducing variables.

1.13.4 CVI interpretation of the online update

The sequential update also admits the approximate-likelihood interpretation

$$q_n(u) \propto \mathcal{N}(Y_{u,n}^e \mid u, V_{u,n}^e) q_{n-1}(u), \quad (100)$$

where the new approximate likelihood term summarises only the information from the current temporal block. Under the Gaussian likelihood,

$$V_{u,n}^e = \sigma^2 (A_n^\top A_n)^{-1}, \quad Y_{u,n}^e = (A_n^\top A_n)^{-1} A_n^\top y_n. \quad (101)$$

Therefore online variational learning can be interpreted as recursively refining the inducing posterior by assimilating one temporal approximate-likelihood factor at a time. This is precisely the online analogue of the static CVI viewpoint introduced earlier, except that the approximate-likelihood contribution is now attached to one temporal block at a time.

1.13.5 Filtering interpretation with fixed spatial inducing points

The above sequential posterior update is also consistent with a filtering view of the temporal inducing state. In particular, the separable construction suggests that the state-space matrices inherit a Kronecker structure of the form

$$A_n^{(st)} = I_{M_s} \otimes A_n^{(t)}, \quad Q_n^{(st)} = K_{ZZ}^{(s)} \otimes Q_n^{(t)}, \quad H^{(st)} = I_{M_s} \otimes H^{(t)}. \quad (102)$$

Hence the spatial inducing locations are fixed across time, while the temporal state is propagated sequentially. This gives a principled online update rule for the spatio-temporal GP in which only the temporal component is updated when new data arrive.

Equation (102) should be understood as a structural interpretation of the online recursion: the temporal dynamics are responsible for state propagation, whereas the spatial covariance enters as a fixed Kronecker factor. This is directly aligned with the modelling assumption that the spatial inducing points are shared across the temporal domain.

1.13.6 Implementation remark

In practice, the online algorithm maintains only the posterior summary (m_n, S_n) , or equivalently (h_n, Λ_n) , together with the fixed spatial matrices $K_{ZZ}^{(s)}$ and $K_{XZ}^{(s)}$. Upon arrival of a new temporal block, one computes the temporal quantities $K_{fu,n}^{(t)}$ and $K_{uu,n}^{(t)}$, forms A_n via (89), updates the sufficient statistics through (96), and then recovers the new posterior moments through (97). This yields an online spatio-temporal variational update rule without reprocessing the full historical dataset.

Equivalently, the implementation may be summarised as follows:

1. keep the spatial inducing locations and spatial covariance factors fixed;
2. update the temporal cross-covariance and temporal inducing covariance for the newly arrived time block;
3. form the Kronecker-separable projection matrix A_n ;
4. update the posterior summary via (96).

This makes explicit how the online recursion is realised in the present Kronecker-separable HiPPO-SVGP framework.

1.14 Predictive Distribution

For a new spatio-temporal input (t_*, s_*) , define the mixed point-to-inducing covariance

$$K_{*u}^{(st)} = k_{fu}^{(t)}(t_*; t) \otimes k_{sZ}^{(s)}(s_*), \quad (103)$$

where

$$k_{sZ}^{(s)}(s_*) = (k_s(s_*, z_1^{(s)}), \dots, k_s(s_*, z_{M_s}^{(s)})). \quad (104)$$

When no ambiguity arises, we write K_{*u} in place of $K_{*u}^{(st)}$.

Then the predictive distribution is Gaussian,

$$q(f(t_*, s_*)) = \mathcal{N}(\mu_*, \sigma_*^2), \quad (105)$$

with predictive mean

$$\mu_* = K_{*u} K_{uu}^{-1} m, \quad (106)$$

and predictive variance

$$\sigma_*^2 = k_t(t_*, t_*) k_s(s_*, s_*) - K_{*u} K_{uu}^{-1} K_{u*} + K_{*u} K_{uu}^{-1} S K_{uu}^{-1} K_{u*}. \quad (107)$$

The three terms in (107) correspond respectively to the prior variance, the variance reduction due to conditioning on the inducing variables, and the posterior uncertainty carried by the variational distribution $q(u)$.

1.15 Kronecker Structure and Computational Remarks

The proposed model exploits two layers of structure:

1. **Temporal analytic structure:** $K_{uu}^{(t)}$ and $K_{fu}^{(t)}$ are available in closed form via the HiPPO-RFF spherical-Bessel construction.
2. **Spatio-temporal Kronecker structure:**

$$K_{uu} = K_{uu}^{(t)} \otimes K_{ZZ}^{(s)}, \quad K_{fu} = K_{fu}^{(t)} \otimes K_{XZ}^{(s)}.$$

For notational simplicity, these are often written as K_{uu} and K_{fu} when no ambiguity arises.

Consequently,

$$K_{uu}^{-1} = (K_{uu}^{(t)})^{-1} \otimes (K_{ZZ}^{(s)})^{-1}, \quad (108)$$

and

$$\log |K_{uu}| = M_s \log |K_{uu}^{(t)}| + M_t \log |K_{ZZ}^{(s)}|. \quad (109)$$

These identities avoid direct dense inversion or Cholesky factorization of an $(M_t M_s) \times (M_t M_s)$ covariance matrix and provide the main computational advantage of the proposed framework.

In practice, the Kronecker structure also enables an implementation in which temporal feature matrices are constructed locally for each batch, while the spatial inducing set Z_s and the spatial covariance factors are shared globally across batches. This yields a natural training strategy with mini-batching over time and, when desired, over space as well.

1.16 Remark

The decomposition

$$\Sigma_{f|u} = K_t \otimes R_s + R_t \otimes P_s \quad (110)$$

reveals a clean interpretation of the approximation error: the first term corresponds to the spatial sparse approximation error, whereas the second term corresponds to the temporal

HiPPO-RFF approximation error. This separation is particularly useful when analyzing the effect of the number of temporal basis functions M_t , the number of random Fourier features R , and the number of spatial inducing points M_s .